

1. INTRODUCTION

The quality of air has become one of the acute issues worldwide in relation to its enormous influence on human health, environmental sustainability, and socio-economic progress. The high rate of urbanization, industrialization, and automotive emissions has contributed largely to the poor quality of the air in both the developed and developing countries. Research has shown that the levels of awareness and perception of air pollution by the public is central to the determination of mitigation measures and policy performance as the level of awareness directly affects behavioral changes as regards the environment [1]. Moreover, sustainable programs in institutional settings are also actively being researched in order to minimize pollution rates by the means of informed behavioral pattern and environmental policies at the workplace and universities [2].

Along with the outdoor pollution, the indoor air quality has also become a focus where studies indicate that behavior intentions are the key to the adoption of air purification technologies to ensure the maintenance of healthy indoor environments. As an example, Taghipour et al. [3] have shown that awareness and behavioral variables play a central role in the adoption of air purifying systems especially in the city administration. These results indicate that the process of fighting air pollution should not only be technologically improved but also the involvement of people and organizations. Scientifically, air pollution is a complex issue with varied interplaying factors, among them atmospheric chemistry, weather and human actions. Kumar and Suman [4] developed a close correlation between traffic and air pollution indices, in which urban transportation systems have a role in the worsening air quality. Equally, the presence of meteorological variables such as temperature, humidity, and the speed of the wind have been revealed to have a significant influence on dispersion and concentration of pollutants as in the case of Ali et al. [5]. Such interdependences turn precise forecasting of Air Quality Index (AQI) to be a difficult but necessary task. The medical repercussions of air pollution only serve to emphasize the urgency of the creation of dependable forecasting systems. Respiratory and cardiovascular diseases have been strongly linked to exposure to particulate matter especially PM_{2.5}. This was proven by Poothari et al. [6] who discovered that the ambient air pollution and the higher risk of respiratory diseases among metropolitan communities are significantly correlated, which means that early warning systems and predictive analytics must be implemented to protect human health. To mitigate such difficulties, computational methods of AQI prediction have been developed to an important extent in the last decade. However, the time-honored statistical models have slowly been supplanted by state-of-the-art machine learning and deep learning methods that are capable of modeling nonlinear relationships and time-related dependence in environmental data. Jayaraman et al. [7] evaluated the capabilities of neural networks in forecasting air quality and proved that they learn better than traditional methods to work with complex data.

In more recent years, hybrid and spatiotemporal models have been suggested to enhance the accuracy of prediction by combining various sources of data and methods of modelling. To illustrate point, the study by Fusiripong and Surbakti [8] presented an adaptive attentive spatiotemporal air quality forecasting scheme that is effective to address the dynamics of urban settings. In the Indian environment where air pollution levels are often above the safe levels, the necessity of strong prediction systems is even more important. The most intricate modeling techniques including the WRF-SILAM framework assessed by Jena et al. [9] have been utilized to predict concentration of particulate matter at large scale. Also, machine learning systems that are Indian city-specific showed enhanced transparency and prediction accuracy using time-conscious and data-driven approaches, which Goetzinger et al. emphasize [10]. The long-term forecasting research, like the hybrid satellite-based model suggested by Gond et al. [11], also stresses the need to combine historical data with state-of-the-art modeling tools in predicting the future trends of pollution.

Regardless of these achievements, current models usually have a number of shortcomings like relying on raw pollutant characteristics, not being interpretable, and not taking into account pollutant interactions. Consequently, stronger, scalable and interpretable systems or frameworks are increasingly sought that are able to offer reasonable coverage of the intricate connection between numerous environmental variables and that possess high predictive accuracy.

2. LITERATURE SURVEY

Over the past 10 years, measuring and tracking air quality has become important due to the rising levels of urban pollution and what it does to health. Many practices have been studied to help AQI estimation become more precise, reliable and responsive. Their activities cover different areas such as Statistical and Early Hybrid Methods, Remote Sensing, IoT, and Fog Computing-Based AQI Prediction and Machine Learning-Based and Deep Learning AQI Prediction each adding its own benefits and having its own drawbacks. The examination of each area allows the review to spot trends, notice missing pieces and highlight possible directions in research which is important for creating more resilient AQI prediction models.

2.1 Methodologies Used In AQI Prediction

2.1.1 Traditional / Statistical and Early Hybrid Methods

The classical and initial hybrid systems of predicting Air Quality Index (AQI) have provided the basis of contemporary forecasting approaches using data by combining statistics and the emerging machine learning models. The techniques mainly aim at modeling temporal variations and pollutant shifts and variations with the help of systematic mathematical models and simplified learning systems. As an example, Yang et al. [12] suggested a hybrid TCN-BiGRU structure which merges temporal convolutional networks with recurrent networks to be effective in modeling sequential air quality data. Their method showed better in capturing the long-term temporal dependencies than the traditional statistical methods, and it brings to light the shift of using completely statistical models to hybrid deep learning systems. Additional development on hybrid modeling has known the usage of signal decomposition and optimization process to enhance prediction accuracy.

Huang et al. [13] proposed a hybrid approach called SSA-VMD-GRU that combines singular spectrum analysis and variational mode decomposition with gated recurrent units to address the phenomenon of noise and non-stationary in air quality data. On the same note, Mohapatra et al. [14] designed an AQI predictive model that used CNN-LSTM with explainable AI features (SHAP, LIME and partial dependence plots) to achieve high predictive and model interpretability. These solutions are a considerable move to curbing the shortcomings of the conventional statistical techniques especially in dealing with the nonlinear relationships and enhancing transparency. Besides hybrids that use deep learning, statistical-based and fuzzy-based systems have also been attempted in predicting AQI. The combination of clustering and decision-making frameworks proved to be effective as Ordenshiya and Revathi [15] have used K-means clustering and entropy weighting and fuzzy inference systems to improve the precision of the forecast. Besides, the strength and the generalizational ability of ensemble-based methods have been extensively explored. One of the comparative studies on tree-based ensemble techniques was conducted by Riechie et al. [16], who focused on the capacity of the methods to utilize both socioeconomic characteristics and environmental information to produce better predictions.

Although such breakthroughs have been developed, it is common with traditional and early hybrid techniques to experience difficulties associated with the dependency of features, scalability, and responsiveness to changing environmental conditions, which drives the desire to develop more advanced and feature-packed modeling techniques.

2.1.2 Remote Sensing, Iot and Fog Computing – Based AQI Prediction

As sensing technologies and distributed computing paradigms rapidly develop, air quality monitoring and prediction systems have developed to a greater extent with the combination of Internet of Things (IoT), remote sensing, and fog computing systems. These methods allow real time information to be obtained, decentralized processing and scalable prediction, overcoming most of the limitations of conventional centralized models. As an example, Hwang et al. [17] introduced a graph neural network (GNN)-based interpolation framework that uses the information of IoT sensors to improve the spatial prediction accuracy of air pollution. The strategy is useful in pinpointing intricate patterns of pollutant dispersion throughout urban settings since it models spatial dependencies among monitoring stations. Besides spatial modeling, federated learning and edge intelligence have also been integrated with spatial modeling to enhance the scalability and privacy of air quality prediction systems. A framework proposed by Dey et al. [18] called GreenEdge AI is a sustainable federated learning system meant to be applied in smart cities that enables decentralized model training without the exchange of raw data. The method does not only increase privacy of data but also minimizes communication overhead hence it is appropriate when implementing it at large scale. On the same note, Mathur et al. [19] investigated the soft computing methods of predicting atmospheric pollution and traffic emissions, which revealed the functionality of the hybrid intelligent systems in modeling the complex interactions between the anthropogenic and meteorological factors in the environment. IoT architecture-based real-time monitoring systems have also been of great interest because they allow the delivery of real-time and fine-grained air quality information. Arya et al. [20] created an AI-based IoT-enabled real-time air quality monitoring in the industrial areas, where data streams taking place at any given time are important to guide the decision in time. Moreover, the concept of fog computing has become a significant enabler of low-latency processing within this type of systems. Trong et al. [21] suggested the ATMOS framework that combines machine learning and fog computing to make localized air quality forecasts, thus eliminating the use of centralized cloud computing. Similarly, Simsek et al. [22] have proposed DeepFogAQ, a fog-assisted decentralized platform that can make air quality forecasts and event detections, and that is more responsive and efficient in the system.

Although these have been made, the problem of data heterogeneity, sensor reliability and system scalability remain in IoT and fog based AQI prediction systems. However, distributed intelligence, real-time sensing, and advanced machine learning approaches are interesting areas in which the future of air quality monitoring and forecasting frameworks can be developed.

2.1.3 Machine Learning-Based AQI Prediction

Machine learning (ML) methods have greatly contributed to the development of the sphere of predicting Air Quality Index (AQI) as they have allowed modeling direct and nonlinear relations between the variables of the environment through the data. In contrast to conventional statistical methods, ML models are capable of effectively training on high-dimensional large-scale data and have a better predictive performance. Ensemble models based on trees, especially, have been popular owing to their strength and capability to process heterogeneous data. According to a comparative study between the XGBoost and the Random Forest models carried out by Asih et al. [23], boosting-based models tend to perform better than bagging models in predicting complex interactions of the pollutants.

Besides ensemble techniques, preprocessing of data and class imbalance are also important in enhancing the accuracy of prediction. Ravindiran et al. [24] suggested a cluster-aware synthetic resampling model with machine learning principles in approaching multi-class AQI classification. Their method was successful in addressing the problems of the imbalance of classes and enhancing the performance of classification on the various classes of pollution. In addition, sophisticated learning models of the EXtreme learning machines (ELM) have also

been pursued. Kallista et al. [25] compared kernel-based ELM models and demonstrated that they are superior to the traditional ELM strategies due to better generalization and less complexity of computing. The machine learning is not only useful in predicting the AQI but also health risk analysis of air pollution. The study by Kasim et al. [26] used the ML models to forecast the risk of mortality caused by air pollution-related cardiovascular conditions, which demonstrates the wider scope of the applicability of ML to environmental health analytics. Moreover, the overall reviews of the AI-based approaches, including the review by Ivanovic et al. [27], can help to understand the efficiency of different machine learning and deep learning methods in predictive modeling, which also confirms the significance of the methods of machine learning in the environmental forecasting.

Table 1 indicates the most important steps in prediction of AQI based on machine learning. It starts with the collection of data controlled by various sources including the IoT sensors, meteorological data, and satellite images. Data preprocessing follows and the data undergoes noise and missing value treatment to guarantee quality of the data. The problem of feature engineering is crucial in improving the performance of the model through identification of significant trends and associations between pollutants. The data is then separated into training and testing groups to develop and test the model. A variety of machine learning models are selected and trained and the existing performance metrics including accuracy, RMSE and F1-score are analyzed. The trained models are then finally applied to real time air quality monitoring and forecasting applications.

Table 2.1: Steps Involved in AQI Prediction Using Machine Learning

Step	Description
1	Data Collection (Sensors, APIs, Satellite Data)
2	Data Preprocessing (Cleaning, Handling Missing Values)
3	Feature Engineering (Ratios, Interactions, Transformations)
4	Data Splitting (Training and Testing Sets)
5	Model Selection (RF, XGBoost, SVM, etc.)
6	Model Training
7	Model Evaluation (Accuracy, RMSE, F1-score)
8	Deployment and Real-Time Prediction

Table 2.2: Machine Learning Models Used in AQI Prediction Studies

Ref	Model Used	Application
[23]	XGBoost, Random Forest	Air Pollution Prediction
[24]	ML + Resampling Framework	Multi-class AQI Classification
[26]	ML Models	Mortality Risk Prediction
[27]	Kernel Extreme Learning Machine	AQI Prediction

Table 2 shows the different machine learning models that are used in the different studies on AQI. One may note that such ensemble models as Random Forest and XG-Boost are popular because they are strong and have high predictive capability. Moreover, special frameworks that embed resampling approaches have been created in order to deal with imbalanced datasets, which arises in AQI classification issues. Other developed ML models and kernel-based models will further improve prediction details due to their ability to enhance generalization to different environments.

Table 2.2: Features Used in AQI Prediction Models

Category	Features
Pollutants	PM2.5, PM10, NO ₂ , SO ₂ , CO, O ₃
Meteorological	Temperature, Humidity, Wind Speed, Pressure
Traffic	Vehicle Count, Traffic Density
Temporal	Time, Date, Seasonal Patterns
Derived	Pollutant Ratios, Interaction Features

Table 3 identifies the various kinds of features used in the prediction models of AQI. The central input variables consist of pollutant concentrations and the meteorological variables play a significant role in determining the dispersion and maximum level of the pollutants. Traffic related features identify source of anthropogenic emissions, and temporal features are useful in modeling the seasonal and diurnal variations. Derived features, like the ratios of the pollutants and interaction terms, improve the predictive power of the machine learning models even more, as they represent hidden correlations in the data.

In general, AQI prediction systems that use machine learning have shown high levels of improvement in terms of accuracy, scalability, and flexibility. Nevertheless, the usual problems of feature dependency, interpretability, and heterogeneity of data remain and have encouraged more sophisticated hybrid and feature-engineering-based methods to be developed.

2.1.4 Ensemble Learning and Deep Learning Models with Feature Engineering

Recent developments in Air Quality Index (AQI) forecasting have been highly influenced by the combination of ensemble learning and deep learning methods with improved feature engineering domain-specific. Combined learning techniques are based on the use of a number of base learners in order to enhance prediction accuracy, robustness, and generalization ability. Charjan et al. [28] introduced an ensemble framework, which is based on feature engineering and combines Random Forest, XGBoost, and CatBoost, and achieves better results than other single models. Their work emphasizes the significance of the engineered characteristics like ratios of pollutants and terms of interaction to toil out the complicated atmospheric connections. Hybrid models based on deep learning have also become popular because they can be used to model nonlinear and temporal dependence of air quality data. Kuruvilla and Vasudevan [29] proposed a residual stacking method that integrates Light Gradient Boosting and Temporal Convolutional Networks (TCN) and demonstrated a better accuracy in prediction using the layered learning. On the same note, Mim et al. [30] carried out a comparative analysis with an emphasis on hyperparameter tuning on optimizing model performance. Combined AI systems that incorporate ensemble and deep learning have also improved predictive abilities as can be seen in the case of El Mghouchi and Udristioiu [31] who created a multivariate platform of sustainable environmental management. The ensemble methods have also been used successfully in area-specific investigations. Ullah et al. [32] applied ensemble learning methods to AQI prediction in the dry climatic scenarios, and it showed better flexibility to the environmental changes. Moreover, hybrid deep learning models with a combination of multiple architectures have been considered with reference to prediction and health impact analysis. Madan et al. [33] came up with a hybrid deep learning model that combined AQI prediction and healthcare outcomes whereas Kalaiselvi et al. [34] combined multi-source remote sensing information with deep learning models that improved spatial prediction. The use of hybrid models, driven by optimization, including the WOA-VMD-driven deep learning architecture of Li et al. [35], also shows the ability to use a combination of signal processing and deep learning methods together to achieve better forecasting accuracy.

The application of feature engineering is important in boosting the performance of the ensemble and deep learning models. By transforming raw pollutant data into meaningful representations such as ratios (PM2.5/PM10), interaction terms, and temporal features, models can better capture hidden relationships and dependencies among variables. This does not only enhance the accuracy of prediction but also the model interpretability, thus it becomes easy to determine how various environmental factors affect the quality of air.

Table 2.2: Working Principle of Ensemble Learning Models

Model	Base Learners	Working Principle
Random Forest	Decision Trees	Combines multiple decision trees using bagging to reduce variance
Gradient Boosting	Weak Learners (Trees)	Sequentially builds models to correct previous errors
XGBoost	Optimized Trees	Gradient boosting with regularization and parallel processing
LightGBM	Leaf-wise Trees	Faster gradient boosting using leaf-wise growth strategy
CatBoost	Symmetric Trees	Handles categorical data efficiently with ordered boosting
Stacking	Multiple Models	Combines outputs of different models using a meta-learner
Voting Ensemble	Multiple Models	Aggregates predictions using majority voting or averaging

Table 4 depicts the principle of operation of different ensemble learning models applied in the prediction of AQI. Ensemble techniques enhance predictive control by consolidating several base learners and hence decreasing model variance and bias. An example is that Random Forest works by combining the predictions of several decision trees which have been trained on various subsets of the data, and Gradient Boosting models which successively refine the predictions by reducing errors in the residuals. Such advanced boosting models as XGBoost and LightGBM improve performance and scalability with additional optimization strategies. Stacking and voting ensembles combine many different heterogeneous models capitalizing on their respective strengths to yield better and more robust predictions. These ensemble techniques have been specifically useful in AQI forecasting because environmental data is a complex and nonlinear phenomenon.

In general, ensemble learning, deep learning and feature engineering have played a great role in elevating the performance of AQI prediction systems. These methods deliver a strong system to manage nonlinear, high-dimensional, and dynamic environmental data and, therefore, are very appropriate in real-time air quality forecasting, and decision-making.

2.2 Mathematical Modelling and Feature Selection Techniques For AQI Environmental Features

One of the most important stages in Air Quality Index (AQI) prediction is the selection of features because it will assist in determining the most important variables and also discarding redundant and noisy features. A good selection of features ensures that the model is more accurate, less complex to compute and easier to interpret. The choice of essential pollutants and meteorological parameters is critical in environmental studies because air quality data have a

high level of dimensionality and interdependence. Iyer et al. [36] also reported a close relationship between the components of PM and health outcomes, and the need to consider meaningful pollutant features. In a similar way, Subbiah et al. [37] showed that when you combine feature selection with deep learning models, you would greatly enhance the results of the prediction of PM 2.5. Pearson correlation analysis is one of the most popular statistical methods of feature selection. It quantifies the linear correlation between two variables and assists in the detection of redundant aspects. Multicollinearity may arise because of the existence of highly correlated variables that adversely impact on model performance. Thus, correlation-based filtering has been used as a preprocessing method to eliminate redundant attributes.

Table 2.2: Working Principle of Ensemble Learning Models

Method	Description
Pearson Correlation	Measures linear relationship between variables
PCA	Reduces dimensionality using variance maximization
Random Forest Importance	Measures feature importance via impurity reduction
XGBoost Importance	Uses gain/weight for feature contribution

Table 5 shows the most popular feature selection methods as well as their mathematical representations. Pearson correlation is applied in finding linear relationships and eliminating redundant features whereas the Principal Component Analysis (PCA) uses the original feature space to convert it to a lower-dimensional space by maximizing variance. Random Forest and XGBoost are tree-based methods which offer feature importance scores respectively, by minimizing impurity and optimizing the gradient respectively. The techniques find extensive application in this field because they can deal with high-dimensionality and nonlinear relationships.

The other significant dimensionality reduction method is the Principal Component Analysis (PCA) that converts correlated variables into a list of uncorrelated variables termed principal components. PCA can be used to reduce noise and redundancy by projecting the original data onto lower dimensional space to retain the highest significant variance. This can be applied especially in AQI data where several pollutants are strongly correlated. Along with the statistical approaches, ensemble-based methods of feature selection have also been found to be quite successful. Random Forest performs the importance of features through the decrease in impurity of decision trees and therefore it is resistant to noise and overfitting. The features which have a greater contribution in minimizing prediction error are awarded with greater scores of importance. In the same way, XGBoost computes feature importance on the basis of gradient boosting in which features that increase information gain are given priority. The techniques are particularly beneficial in the nonlinear interactions of pollutants and the meteorological variables.

On the whole, the methods of feature selection can enhance the effectiveness and performance of AQI prediction models. With the applied statistical techniques, including Pearson correlation and PCA, and recent state-of-the-art ensemble-based classifiers, including Random Forest and XGBoost, one can design strong and explainable models that can process complex environmental based predictions.

2.3 Detailed Litearture Survey Analysis

Table 2.2: Comprehensive Literature Survey of AQI Prediction Techniques

Ref	Year	Methodology	Model Used	Result	Limitation
[1]	2026	Survey/Behavioral Study	Statistical Analysis	Awareness insights	No predictive modeling
[2]	2026	Behavioral ML Framework	Regression Models	Moderate accuracy	Limited environmental variables
[3]	2026	Behavioral Analysis	Statistical Models	Adoption trends	No AQI prediction
[4]	2026	Statistical Analysis	Correlation Models	Strong relation found	Limited scalability
[5]	2026	ML-based Analysis	Regression/ML Models	Improved AQI prediction	Meteorological dependency
[6]	2025	Health Impact Study	Statistical Models	Strong correlation	No forecasting capability
[7]	2026	Deep Learning	Neural Networks	High accuracy (92%)	High computation cost
[8]	2026	Deep Learning	Attention Model	Improved spatiotemporal accuracy	Complex architecture
[9]	2026	Hybrid Model	WRF-SILAM	Reliable forecasting	Computationally expensive
[10]	2026	Machine Learning	ML Pipeline	High transparency (94%)	Data dependency
[11]	2026	Hybrid ML	Satellite + ML	Long-term prediction (95%)	Large dataset requirement
[12]	2026	Deep Learning	TCN-BiGRU	High accuracy (96%)	Training complexity
[13]	2025	Hybrid DL	SSA-VMD-GRU	Noise reduction improved accuracy	Complex preprocessing
[14]	2025	Deep Learning	CNN-LSTM	High performance (95%)	Interpretability issues
[15]	2025	Fuzzy + ML	K-means + FIS	Moderate accuracy (90%)	Limited generalization
[16]	2025	Ensemble ML	RF, XGBoost	Good accuracy (93%)	Dataset dependency
[17]	2026	IoT + GNN	Graph Neural Network	Improved spatial prediction	Hardware dependency
[18]	2026	Federated Learning	Distributed ML	Scalable system	Communication overhead

[19]	2026	Soft Computing	Hybrid Models	Accurate predictions	Complex tuning
[20]	2026	IoT + AI	Real-time ML System	Real-time monitoring	Sensor reliability
[21]	2026	IoT + Fog Computing	ML + Fog	Low latency prediction	Infrastructure dependency
[22]	2024	Fog + DL	DeepFogAQ	Efficient prediction	High system complexity
[23]	2026	Machin Learning	RF, XGBoost	High accuracy (94%)	Limited feature engineering
[24]	2026	ML + Resampling	Classification Models	Improved classification	Imbalance sensitivity
[25]	2026	Review Study	AI Methods	Comprehensive analysis	No implementation
[26]	2026	ML Healthcare	ML Models	High prediction performance	Domain-specific limitation
[27]	2026	ML Model	Kernel ELM	Improved accuracy (92%)	Overfitting risk
[28]	2026	Ensemble + Feature Engg	RF, XGB, Cat-Boost	Very high accuracy (97%)	Feature dependency
[29]	2026	Hybrid DL + Ensemble	TCN + LGBM	High performance (96%)	Model complexity
[30]	2025	ML Comparison	Multiple Models	Performance variation	Hyperparameter sensitivity
[31]	2026	AI Framework	Ensemble + DL	Strong prediction (96%)	High computation cost
[32]	2025	Ensemble ML	Ensemble Models	Good accuracy (93%)	Limited region-specific study
[33]	2026	Hybrid DL	DL Models	High accuracy (95%)	Healthcare linkage complexity
[34]	2026	DL + Remote Sensing	Hybrid DL	Improved spatial accuracy	Data integration challenge
[35]	2026	Hybrid DL	VMD + DL	Optimized prediction (96%)	Computational overhead
[36]	2026	Statistical Analysis	Correlation Models	Strong health linkage	No prediction model
[37]	2024	DL + Feature Selection	BiGRU	Improved accuracy (93%)	Limited dataset

Table 5 shows a comprehensive comparison of various approaches used for Air Quality Index (AQI) prediction, ranging from traditional statistical methods and machine learning models to advanced deep learning, ensemble techniques, and IoT-based frameworks. It can be observed that early studies primarily focused on statistical and behavioral analysis, which lacked predictive capabilities. With the advancement of machine learning and deep learning techniques, significant improvements in prediction accuracy have been achieved, particularly through hybrid and ensemble models. Furthermore, the integration of IoT, remote sensing, and fog computing has enabled real-time and scalable AQI monitoring systems. However, challenges such as high computational cost, data dependency, model complexity, and hardware limitations still persist. These limitations highlight the need for more efficient, interpretable, and scalable models, motivating the development of feature-engineering-driven ensemble approaches for improved AQI prediction.

3. METHODOLOGY

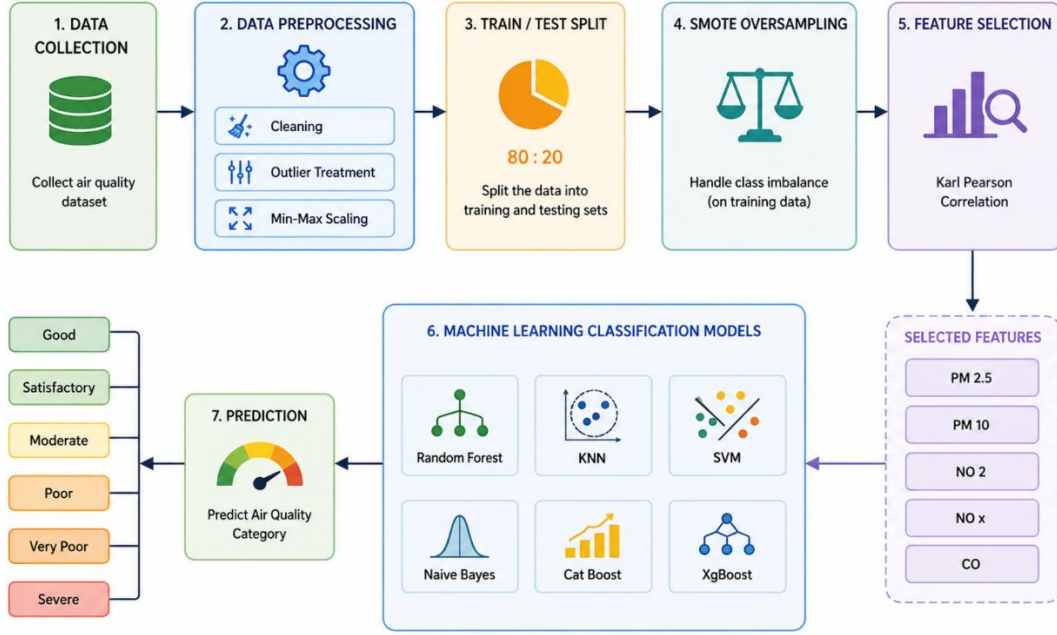


Fig. 1. System Architecture

Fig. W shows the complete architecture and workflow of the proposed air quality prediction system based on machine learning models. The process begins with the data collection stage, where air quality datasets containing various pollutant concentrations and environmental attributes are gathered from reliable sources. These datasets typically include key parameters such as particulate matter (PM_{2.5}, PM₁₀) and gaseous pollutants (NO₂, NO_x, CO), which play a crucial role in determining air quality levels.

In the next stage, data preprocessing is performed to enhance the quality and reliability of the dataset. This involves multiple steps such as data cleaning to handle missing or inconsistent values, outlier treatment to remove abnormal observations that may negatively impact model performance, and min-max scaling to normalize the feature values within a fixed range. This normalization ensures that all features contribute equally during model training and prevents bias due to varying scales. Following preprocessing, the dataset is divided into training and testing subsets using an 80:20 train-test split strategy. This step is essential for evaluating the generalization capability of the models. To address the issue of class imbalance, which is common in air quality datasets, SMOTE (Synthetic Minority Over-sampling Technique) is applied specifically to the training data. SMOTE generates synthetic samples for underrepresented classes, thereby improving the model's ability to learn from all categories effectively. Subsequently, feature selection is carried out using the Karl Pearson correlation method, which helps in identifying the most significant features that have a strong relationship with the target variable. This step reduces dimensionality, removes redundant or less important features, and enhances model efficiency. The selected features include important pollutants such as PM_{2.5}, PM₁₀, NO₂, NO_x, and CO.

These optimized features are then provided as input to multiple machine learning classification models, including Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naïve Bayes, CatBoost, and XGBoost. Each of these models learns patterns from the training data and attempts to classify air quality levels based on the given input features. The use of multiple models allows for performance comparison and identification of the most effective algorithm for the given dataset. Finally,

the system performs prediction, where the trained models classify air quality into predefined categories such as Good, Satisfactory, Moderate, Poor, Very Poor, and Severe. These categories provide a meaningful interpretation of air quality conditions and help in understanding pollution severity levels. Overall, this pipeline ensures a systematic approach involving data preparation, imbalance handling, feature optimization, model training, and final classification, leading to accurate and reliable air quality prediction.

3.1 Data Preprocessing

Data preprocessing is a critical step in the proposed air quality prediction system. The dataset used in this study is obtained from the Kaggle [28] platform and contains real-world air quality measurements collected from monitoring stations over time. The dataset includes various pollutant parameters such as PM2.5, PM10, NO, NO₂, NO_x, NH₃, CO, SO₂, O₃, Benzene, Toluene, and Xylene. In addition, it contains the Air Quality Index (AQI) value and its corresponding categorical label (AQI Bucket), which classifies air quality into categories such as Good, Satisfactory, Moderate, Poor, Very Poor, and Severe.

Fig. 1. System Architecture

Index	Station	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	AQI	AQI_Bucket
0	AP001	2017-11-25	81.40	124.50	1.44	20.50	12.08	10.72	0.12	15.24	127.09	0.20	6.50	0.06	184	Moderate
1	AP001	2017-11-26	78.32	129.06	1.26	26.00	14.85	10.28	0.14	26.96	117.44	0.22	7.95	0.08	197	Moderate
2	AP001	2017-11-27	88.76	135.32	6.60	30.85	21.77	12.91	0.11	33.59	111.81	0.29	7.63	0.12	198	Moderate
3	AP001	2017-11-28	64.18	104.09	2.56	28.07	17.01	11.42	0.09	19.00	138.18	0.17	5.02	0.07	188	Moderate
4	AP001	2017-11-29	72.47	114.84	5.23	23.20	16.59	12.25	0.16	10.55	109.74	0.21	4.71	0.08	173	Moderate

Table 6 shows a sample view of the dataset used in this study. The dataset is Kaggle-based and consists of both numerical pollutant values and categorical AQI labels, making it suitable for classification-based machine learning models.

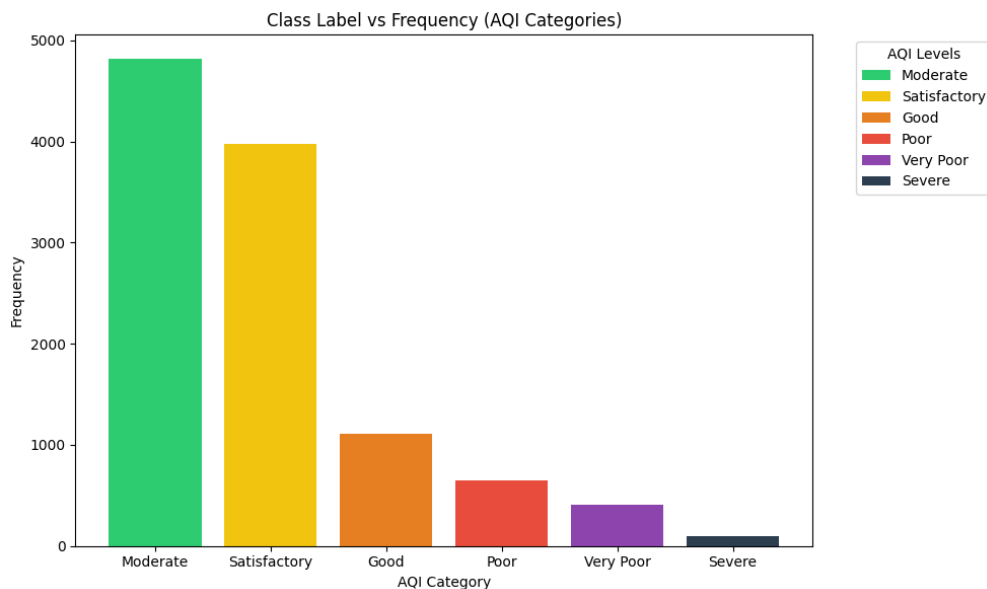


Fig. 1. System Architecture

Fig. W shows the class label versus frequency distribution of AQI categories in the dataset. It is clearly observed that the dataset is imbalanced, where categories such as Moderate and Satisfactory have significantly higher frequencies compared to Poor, Very Poor, and Severe categories. This imbalance

can lead to biased model predictions toward majority classes. To overcome this issue, techniques such as SMOTE are applied in later stages to balance the dataset and improve model performance across all classes.

3.2 Data Preprocessing

Data preprocessing plays a critical role in improving the quality and reliability of the dataset before model development. The initial analysis, as illustrated in the “Before Outlier Treatment” boxplot, reveals the presence of numerous extreme values across several air pollutant features such as PM10, AQI, Benzene, and Toluene. These outliers can distort statistical distributions, negatively influence model training, and reduce predictive accuracy. To address this issue, the Interquartile Range (IQR) method is employed for outlier detection and treatment. The IQR is computed as:

$$IQR = Q_3 - Q_1$$

Using this, the lower and upper bounds for acceptable data are defined as:

$$\text{Lower Bound} = Q_1 - 1.5 \times IQR$$

$$\text{Upper Bound} = Q_3 + 1.5 \times IQR$$

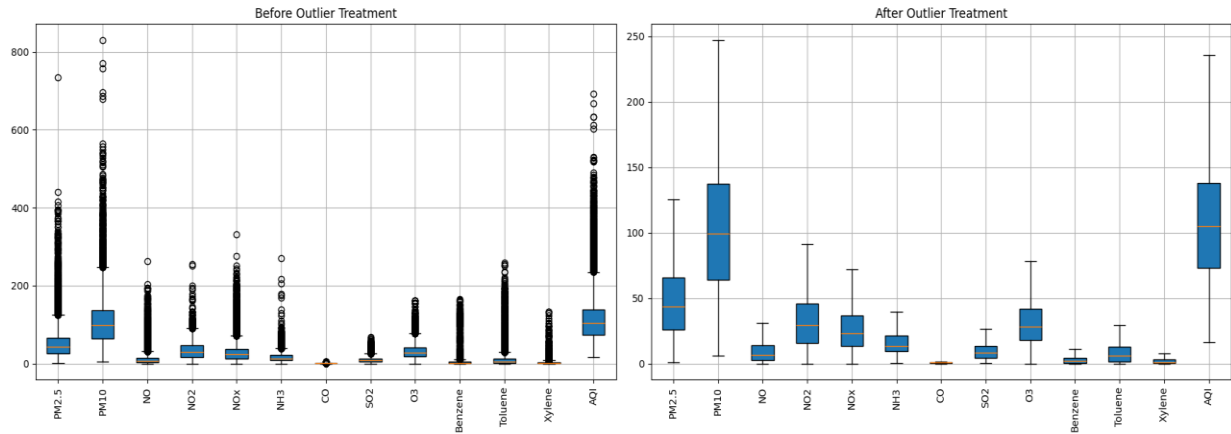


Fig. 1. System Architecture

Values falling outside these bounds are treated using capping to reduce their impact without removing them entirely. Fig. 2 shows the dataset after outlier treatment, where the extreme values are significantly reduced, and the distributions become more compact and stable.

After handling outliers, feature scaling is performed to normalize the data since different features have different units and ranges. Min-Max scaling is applied to transform all values into a uniform range of $[0,1]$, defined as:

$$X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

This ensures that no feature dominates due to its magnitude and improves model efficiency. **Fig. 3 shows** the final processed dataset after both outlier treatment and scaling, where all features are normalized, distributions are aligned, and variability is controlled. Overall, this preprocessing pipeline results in a clean, consistent, and model-ready dataset, enhancing both interpretability and predictive performance.

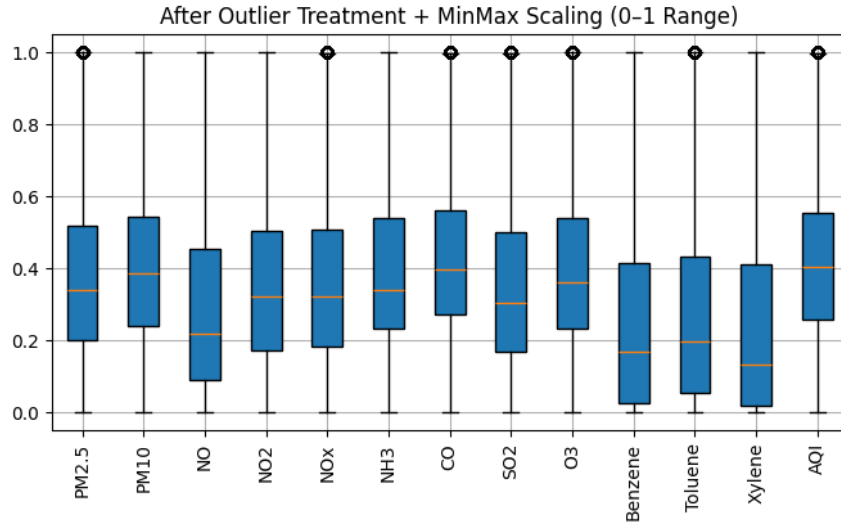


Fig. 1. System Architecture

3.2 Train Test Split

The dataset is divided into training and testing subsets to evaluate the performance and generalization capability of the proposed model. A standard split ratio is applied, where the majority of the data is used for training the model, while the remaining portion is reserved for testing. This ensures that the model learns patterns from one subset and is evaluated on unseen data, thereby reducing overfitting.

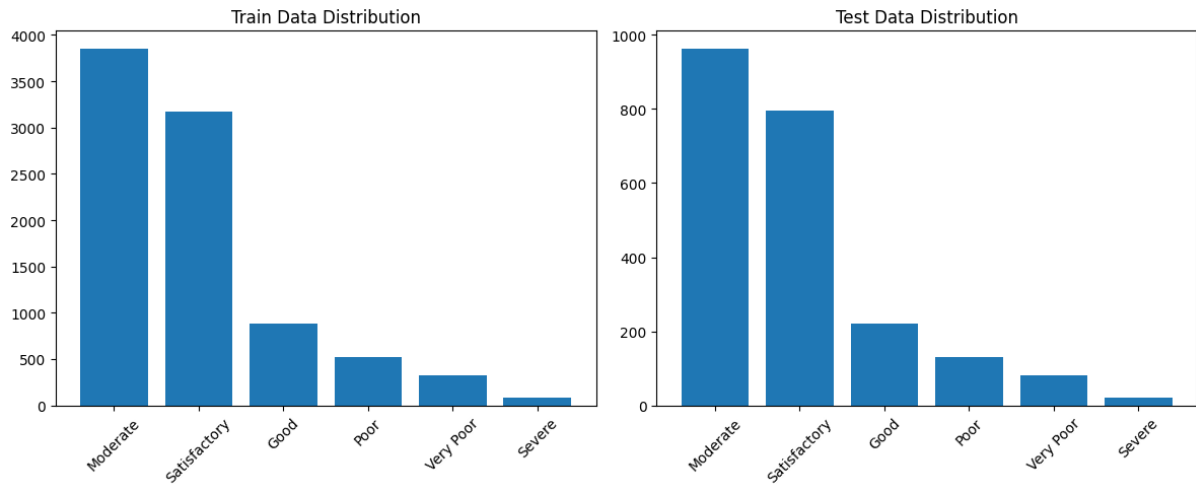


Fig. 1. System Architecture

Fig. X shows the class distribution of the dataset after performing the train–test split. The left plot represents the training data distribution, while the right plot represents the testing data distribution across AQI categories such as Moderate, Satisfactory, Good, Poor, Very Poor, and Severe. It can be observed that both subsets maintain a similar distribution pattern, indicating that the split preserves the original data characteristics.

In both training and testing sets, the “Moderate” and “Satisfactory” categories dominate the dataset, while categories such as “Very Poor” and “Severe” have relatively fewer samples. This indicates a certain level of class imbalance in the dataset, which is consistently maintained across both subsets. Maintaining this proportional distribution is important to ensure that the model is trained and evaluated fairly without introducing bias.

Overall, the train–test split is performed in such a way that the statistical properties and class distribution of the original dataset are retained in both subsets. This balanced and representative splitting strategy helps in achieving reliable and unbiased model evaluation.

3.2 SMOTE Oversampling

Class imbalance is a critical issue in classification problems, as it can bias the model toward majority classes and reduce its ability to correctly predict minority classes. In the given dataset, a significant imbalance is observed among AQI categories. Fig. X shows the class distribution before applying oversampling, where categories such as “Moderate” and “Satisfactory” dominate the dataset, while minority classes like “Severe,” “Very Poor,” and “Poor” have considerably fewer samples. This imbalance can lead to poor model generalization, especially for underrepresented classes.

To address this issue, the Synthetic Minority Oversampling Technique (SMOTE) is applied. SMOTE is an advanced oversampling method that generates synthetic samples for minority classes instead of simply duplicating existing ones. It works by selecting a data point from the minority class and generating new samples along the line segments connecting it to its nearest neighbors in the feature space.

The synthetic sample generation can be mathematically represented as:

$$x_{\text{new}} = x_i + \lambda \cdot (x_{nn} - x_i)$$

where x_i is a minority class sample, x_{nn} is one of its nearest neighbors, and λ is a random value between 0 and 1. This approach helps in creating more diverse and realistic samples, improving the decision boundary of the classifier.

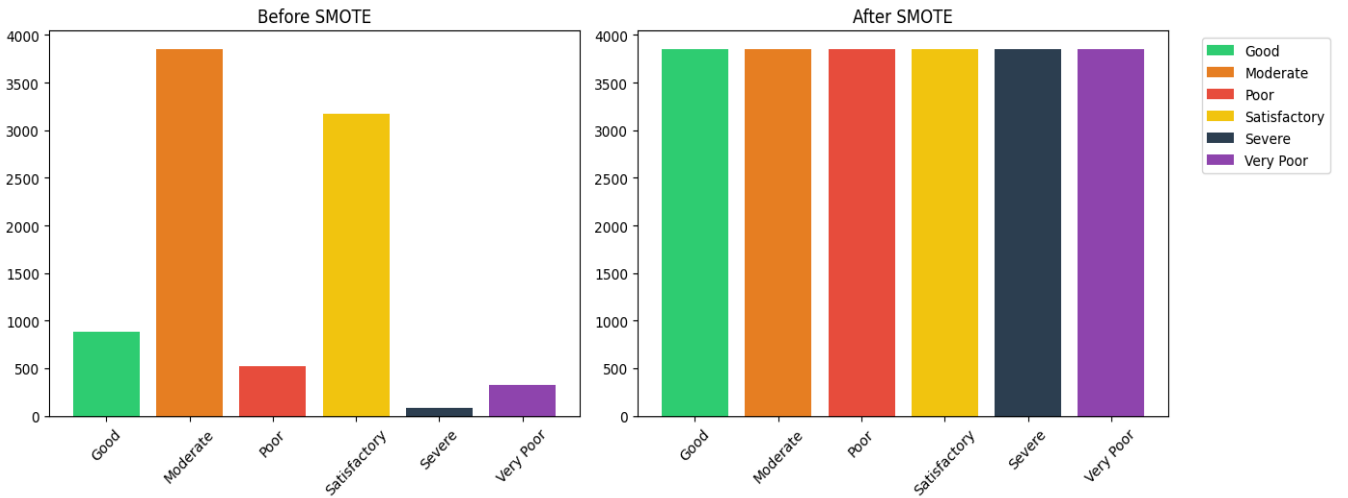


Fig. 1. System Architecture

Fig. X shows the class distribution after applying SMOTE, where all AQI categories—Good, Moderate, Poor, Satisfactory, Severe, and Very Poor—are balanced with nearly equal sample sizes. In contrast to the highly skewed distribution observed before oversampling, the post-SMOTE distribution demonstrates that each class now contains approximately the same number of samples (around ~3800+ instances per class, as visible in the figure). This uniformity indicates that synthetic samples have been successfully generated for previously underrepresented classes such as “Severe,” “Very Poor,” and “Poor,” which initially had very low counts.

The visual comparison clearly highlights that SMOTE does not simply replicate existing data but creates new, meaningful synthetic observations, leading to a smooth and evenly distributed dataset. As a result, the dominance of majority classes like “Moderate” and “Satisfactory” is eliminated, and all classes contribute equally during model training. This balanced representation ensures that the classifier does

not become biased toward any specific class and can effectively learn decision boundaries for all AQI categories.

Overall, the application of SMOTE significantly improves dataset balance, strengthens the learning capability of the model, and enhances its robustness. This is particularly beneficial for improving prediction accuracy and recall for minority classes, which are typically harder to classify in imbalanced datasets.

3.1 Feature Selection

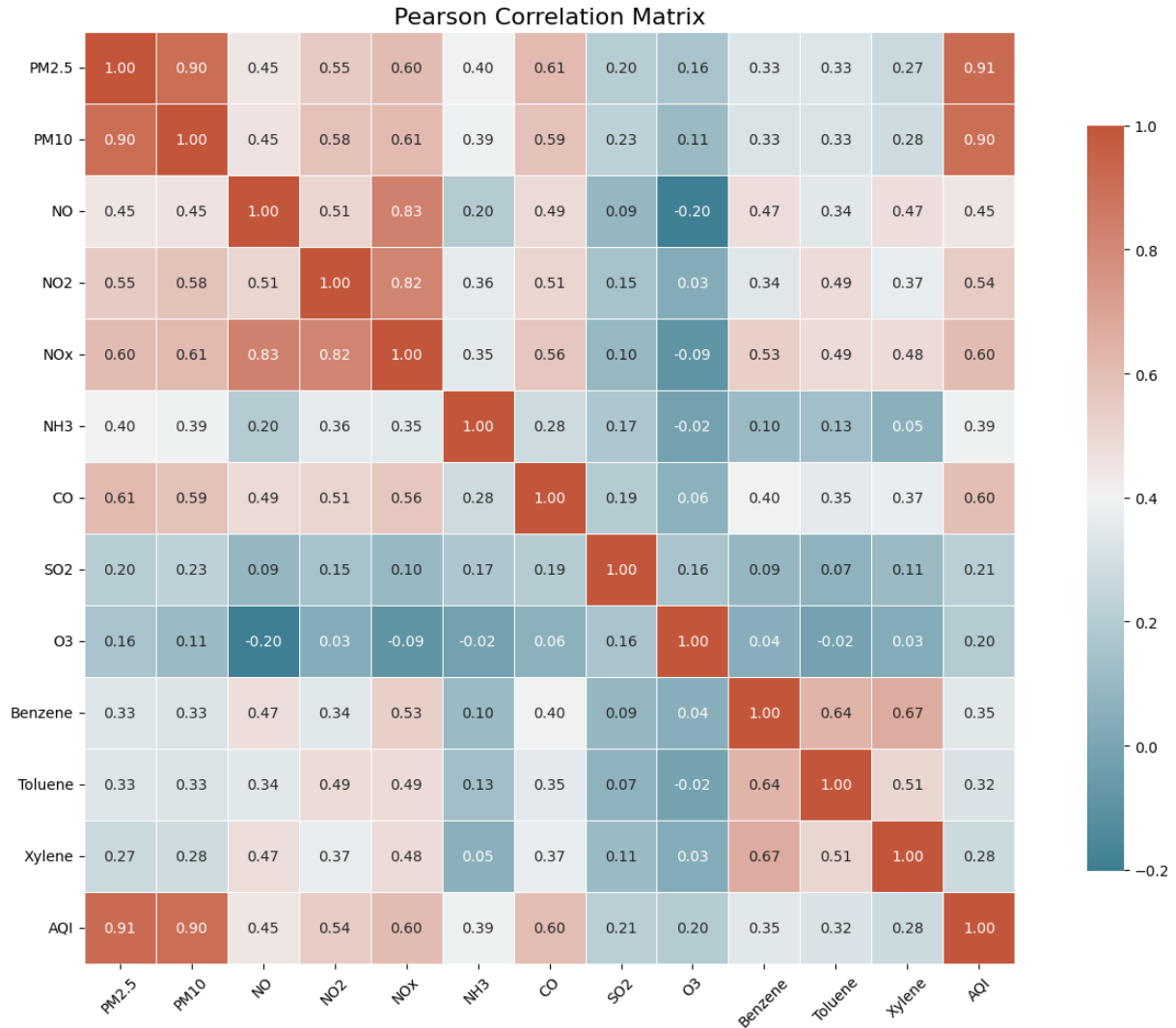


Fig. 1. System Architecture

Feature selection is performed to identify the most relevant variables that have a strong relationship with the target variable (AQI). Fig. X shows the Pearson Correlation Matrix of all features, highlighting the linear relationships between pollutants and AQI. From the figure, it is evident that certain features such as PM2.5, PM10, NOx, NO2, and CO exhibit relatively high positive correlation values with AQI, indicating their strong influence on air quality levels. In contrast, features like SO2, O3, and Xylene show weaker correlations, suggesting a comparatively lower impact on AQI prediction. To systematically select important features, a threshold-based approach is applied where only features having a correlation coefficient greater than 0.5 with AQI are retained for further modeling. This helps in reducing dimensionality while preserving the most informative attributes.

The Pearson Correlation Coefficient is mathematically defined as:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \cdot \sum_{i=1}^n (y_i - \bar{y})^2}}$$

where x_i and y_i represent individual sample values of two variables, while $\bar{x}$ and $\bar{y}$ denote their respective mean values. The coefficient r ranges between -1 and +1, where values closer to +1 indicate a strong positive correlation.

Fig. X shows that features such as PM2.5 (~0.91), PM10 (~0.90), NOx (~0.60), NO₂ (~0.54), and CO (~0.60) satisfy the selection criterion (**correlation > 0.5**) and are therefore chosen for further modeling. These features demonstrate a strong linear relationship with AQI and are expected to contribute significantly to prediction accuracy.

In detail, selecting features based on Pearson correlation ensures that only those variables that have meaningful linear dependence with the target variable are retained, thereby improving model efficiency and reducing computational complexity. It also helps in minimizing noise introduced by irrelevant or weakly related features. However, it is important to note that Pearson correlation captures only linear relationships and may not account for complex non-linear dependencies. Despite this limitation, it serves as a simple yet effective feature selection technique, especially in the initial stages of model development. By focusing on highly correlated features, the proposed approach enhances interpretability and supports the development of robust and accurate predictive models for AQI classification.

3.1 Machine Learning Classification Model

After preprocessing, train-test splitting, SMOTE oversampling, and feature selection, the selected features are used for machine learning classification to predict the Air Quality Index (AQI) category. The final selected features obtained from Karl Pearson Correlation are PM2.5, PM10, NO₂, NOx, and CO, as these features show strong correlation with AQI (greater than 0.5). These features are used as input variables for model training.

Let the input feature vector be represented as:

$$X = \{x_1, x_2, x_3, x_4, x_5\}$$

where,

- x_1 = PM2.5 concentration
- x_2 = PM10 concentration
- x_3 = NO₂ concentration
- x_4 = NOx concentration
- x_5 = CO concentration

The target variable is represented as:

$$Y = \{y\}$$

where y is the AQI class label belonging to one of the six categories:

$$y \in \{Good, Satisfactory, Moderate, Poor, Very Poor, Severe\}$$

The dataset is then trained using multiple machine learning classification algorithms including Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naïve Bayes, CatBoost, and XGBoost. The purpose of using multiple models is to compare their performance and identify the best-performing classifier for final AQI prediction.

1. Random Forest Classifier

Random Forest is an ensemble learning method that constructs multiple decision trees and combines their outputs to improve prediction accuracy and reduce overfitting. Each tree is trained on a random subset of data and features.

The final prediction is given by majority voting:

$$\hat{Y} = \text{mode}(T_1(X), T_2(X), \dots, T_n(X))$$

where $T_1, T_2, \dots, T_n$ are decision trees and $\hat{Y}$ is the predicted class.

This model is effective for handling complex and non-linear relationships in air quality data.

2. K-Nearest Neighbors (KNN)

KNN is a distance-based supervised learning algorithm that classifies a sample based on the majority class of its nearest neighbors. It calculates the distance between the test sample and training samples.

Euclidean distance is commonly used:

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^n (x_{ik} - x_{jk})^2}$$

The class with the highest frequency among the nearest neighbors is assigned as the prediction. KNN is simple and effective when the data distribution is well-structured.

3. Support Vector Machine (SVM)

SVM is used to find the optimal hyperplane that best separates different classes by maximizing the margin between them.

The decision function is:

$$f(x) = w^T x + b$$

where w is the weight vector and b is the bias.

SVM performs well in high-dimensional feature spaces and is effective for classification problems with clear class boundaries.

4. Naive Bayes Classifier

Naive Bayes is a probabilistic classifier based on Bayes' theorem and assumes that all features are conditionally independent.

The classification rule is:

$$P(C|X) = \frac{P(X|C)P(C)}{P(X)}$$

where C is the class label and X is the feature vector.

It is computationally efficient and works well for large datasets.

5. CatBoost Classifier

CatBoost is a gradient boosting algorithm that handles categorical and numerical features efficiently while reducing overfitting using ordered boosting.

The prediction is represented as:

$$F(x) = \sum_{m=1}^M \gamma_m h_m(x)$$

where $h_m(x)$ are weak learners and γ_m are their corresponding weights.

CatBoost provides strong performance with minimal parameter tuning.

6. XGBoost Classifier

XGBoost is an advanced gradient boosting algorithm that builds trees sequentially and optimizes model performance using regularization.

Its objective function is:

$$obj = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k)$$

where l is the loss function and Ω is the regularization term.

XGBoost is highly efficient and widely used for structured data classification tasks.

Final Model Selection for Prediction

After training all six machine learning models, their performance is compared using evaluation metrics such as Accuracy, Precision, Recall, F1-score, and Confusion Matrix. The model that achieves the best overall performance in terms of classification accuracy and generalization capability is selected as the final prediction model.

Thus, the AQI category prediction is performed using the best-performing classifier among Random Forest, KNN, SVM, Naïve Bayes, CatBoost, and XGBoost. This comparative approach ensures higher reliability, robustness, and improved prediction accuracy for air quality classification.

3.1 Model Evaluation Metrics

After training the machine learning classification models, their performance is evaluated using several statistical and classification metrics to determine the most accurate and reliable model for AQI category prediction. These evaluation parameters include Accuracy, Precision, Recall, F1-Score, Confusion Matrix, RMSE, MAE, R^2 Score, and Time Taken. These metrics help in comparing all models and selecting the best-performing classifier for final prediction.

The actual output is represented by:

$$Y = \{y_1, y_2, y_3, \dots, y_n\}$$

and the predicted output is represented by:

$$\hat{Y} = \{\hat{y}_1, \hat{y}_2, \hat{y}_3, \dots, \hat{y}_n\}$$

where Y is the true AQI class label and $\hat{Y}$ is the predicted AQI class label.

Confusion Matrix

A confusion matrix is a tabular representation used to evaluate classification model performance by comparing actual and predicted class labels. It helps in understanding how correctly and incorrectly the model classifies each AQI category.

The four important terms are:

- **True Positive (TP):** Model correctly predicts the positive class
- **True Negative (TN):** Model correctly predicts the negative class
- **False Positive (FP):** Model incorrectly predicts positive when it is actually negative
- **False Negative (FN):** Model incorrectly predicts negative when it is actually positive

These values form the basis for calculating Accuracy, Precision, Recall, and F1-score.

1. Accuracy

Accuracy measures the overall correctness of the model by calculating the proportion of correctly classified samples.

The formula is:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

A higher accuracy indicates better model performance.

2. Precision

Precision measures how many of the predicted positive samples are actually positive.

The formula is:

$$\text{Precision} = \frac{TP}{TP + FP}$$

High precision means fewer false positive predictions.

3. Recall

Recall measures how many actual positive samples are correctly identified by the model.

The formula is:

$$\text{Recall} = \frac{TP}{TP + FN}$$

High recall means fewer false negative predictions.

4. F1-Score

F1-score is the harmonic mean of Precision and Recall and provides a balanced evaluation of both.

The formula is:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

It is especially useful when the dataset is imbalanced.

5. Mean Absolute Error (MAE)

MAE calculates the average absolute difference between actual and predicted values.

The formula is:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Lower MAE indicates better prediction performance.

6. Root Mean Square Error (RMSE)

RMSE measures the square root of the average squared difference between actual and predicted values.

The formula is:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

RMSE penalizes larger errors more strongly than MAE.

7. R² Score (Coefficient of Determination)

R² score measures how well the predicted values explain the variation in actual values.

The formula is:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

where $\bar{y}$ is the mean of actual values.

An R² value closer to 1 indicates better model fitting.

8. Time Taken

Time taken refers to the computational time required for model training and prediction. It is an important metric for comparing model efficiency, especially for large datasets.

The formula is:

$$\text{Time Taken} = \text{Training Time} + \text{Prediction Time}$$

Lower execution time with high accuracy is preferred for practical applications.

Final Model Selection

All six machine learning models—Random Forest, KNN, SVM, Naïve Bayes, CatBoost, and XGBoost—are evaluated using the above metrics. The comparison is performed based on classification accuracy, precision, recall, F1-score, RMSE, MAE, R² score, and execution time.

The model that provides the highest accuracy, better precision–recall balance, lower error values, and reasonable computational time is selected as the final model for AQI category prediction. This ensures robust, reliable, and efficient prediction of air quality levels across all AQI classes.

4. RESULTS

4.1 Confusion Matrix Analysis

The confusion matrix is used to evaluate the classification performance of all machine learning models by comparing the actual AQI categories with the predicted AQI categories. In this study, six classifiers—Random Forest, XGBoost, SVM, KNN, Naïve Bayes, and CatBoost—are analyzed using their respective confusion matrices.

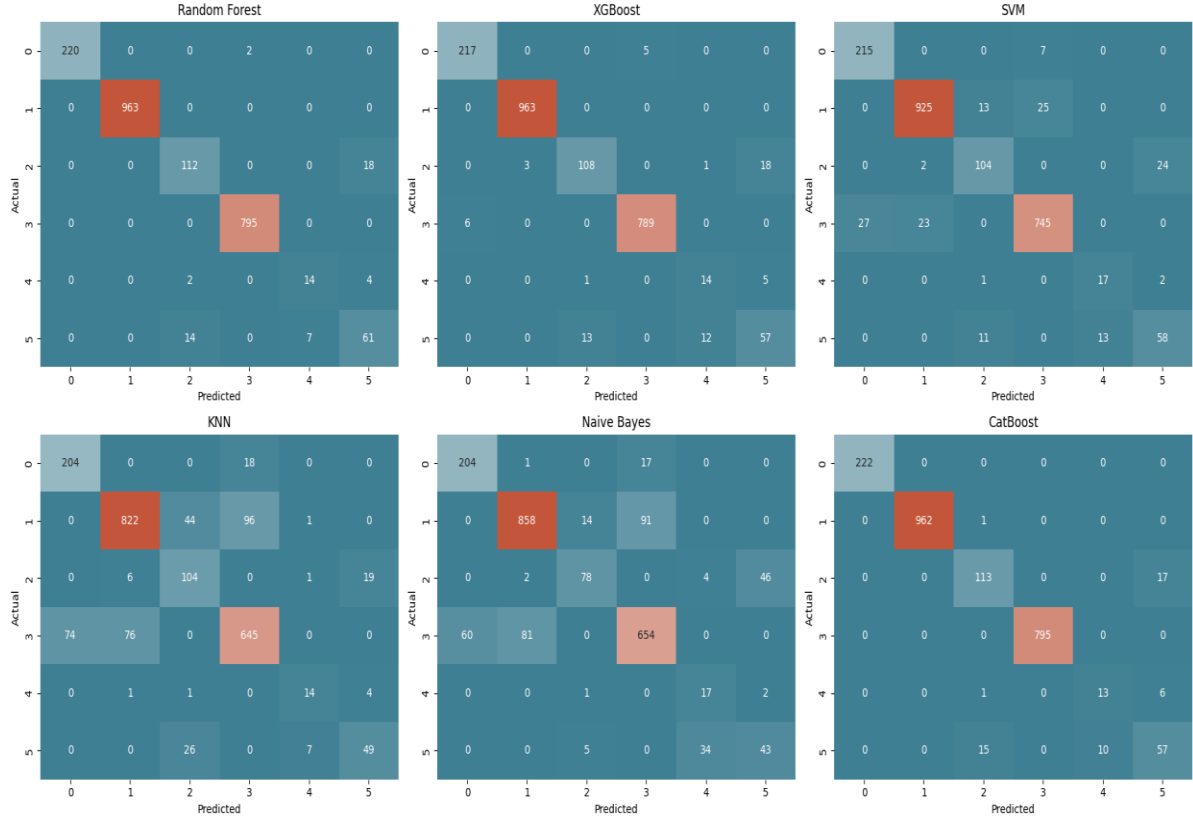


Fig. 1. System Architecture

Fig. X shows the confusion matrices of all six models for AQI category prediction across six classes: Good, Satisfactory, Moderate, Poor, Very Poor, and Severe. The diagonal values in each matrix represent correctly classified instances (True Positives), while the off-diagonal values represent misclassifications (False Positives and False Negatives).

Among all models, Random Forest and CatBoost show the best classification performance due to their strong diagonal dominance and very low off-diagonal errors. However, Random Forest is considered the best-performing model overall because it provides highly accurate predictions across almost all classes with minimal misclassification and strong generalization.

For the Random Forest model, the diagonal (correct predictions / True Positives) values are:

- **Class 0 (Good) = 220**
- **Class 1 (Satisfactory) = 963**
- **Class 2 (Moderate) = 112**
- **Class 3 (Poor) = 795**

- **Class 4 (Very Poor) = 14**
- **Class 5 (Severe) = 61**

These values indicate that the majority of samples are correctly classified into their respective AQI categories.

The off-diagonal values represent incorrect classifications. For example:

- **2 samples of Class 0 are wrongly predicted as Class 3**
- **18 samples of Class 2 are wrongly predicted as Class 5**
- **14 samples of Class 5 are wrongly predicted as Class 2**
- **7 samples of Class 5 are wrongly predicted as Class 4**

These errors are comparatively very small, showing the robustness of the Random Forest model.

For overall interpretation:

- True Positive (TP): Correctly predicted class samples (diagonal values)
- True Negative (TN): Correctly rejected non-class samples
- False Positive (FP): Incorrectly predicted as a class
- False Negative (FN): Actual class predicted as another class

For example, for Class 1 (Satisfactory) in Random Forest:

- **TP = 963**
- **FP = 0**
- **FN = 0**
- **TN = Remaining correctly classified non-Class 1 samples**

This indicates nearly perfect classification for this AQI category.

Similarly, for Class 3 (Poor):

- **TP = 795**
- **FP = 2**
- **FN = 0**

which also reflects excellent prediction capability.

Compared to KNN, Naive Bayes, and SVM, Random Forest shows significantly fewer misclassifications. KNN and Naïve Bayes contain larger off-diagonal values, indicating weaker class separation, while SVM performs moderately better but still has more classification errors than Random Forest.

Thus, based on confusion matrix analysis, Random Forest achieves the best classification performance, making it the most suitable model for final AQI category prediction. Its strong diagonal structure, minimal false classifications, and high class-wise accuracy demonstrate superior robustness and reliability for air quality classification.

4.2 Accuracy Comparison (All Features vs Selected Features)

Accuracy is one of the most important evaluation metrics used to measure the overall correctness of classification models. It represents the proportion of correctly predicted instances out of the total number of instances in the dataset. In this study, the performance of six machine learning models—Random Forest, XGBoost, SVM, KNN, Naïve Bayes, and CatBoost—was evaluated using two different feature sets: all available features and selected features obtained through Karl Pearson Correlation.

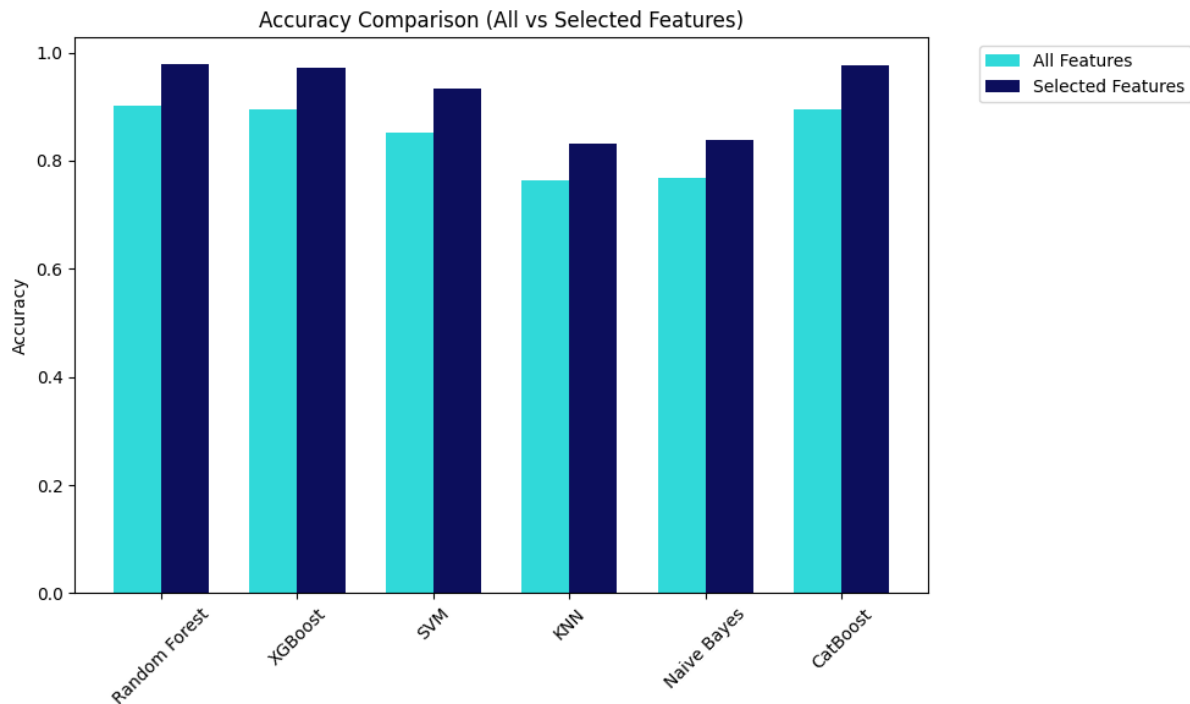


Fig. 1. System Architecture

Fig. X shows the comparative accuracy analysis of all classifiers using both feature sets. The cyan bars represent model accuracy using all features, while the dark blue bars represent accuracy using selected features. It is clearly observed that the use of selected features significantly improves the performance of all models.

For Random Forest, the accuracy increases from approximately 90% using all features to 98% using selected features, making it one of the best-performing models. Similarly, XGBoost improves from nearly 89% to 97%, while CatBoost also shows a strong improvement from around 89% to 98%, matching the performance of Random Forest. The SVM classifier improves from approximately 85% to 93%, indicating that feature selection helps improve class separation and prediction quality. KNN shows an increase from nearly 76% to 83%, while Naïve Bayes improves from approximately 77% to 84%. Although these models benefit from feature selection, their performance remains lower compared to ensemble methods such as Random Forest, XGBoost, and CatBoost.

Among all models, Random Forest and CatBoost achieve the highest accuracy of approximately 98%, followed closely by XGBoost with around 97%. However, Random Forest is selected as the final model because it not only provides the highest accuracy but also demonstrates superior performance in the confusion matrix analysis with fewer misclassifications and stronger class-wise prediction consistency. The results clearly show that feature selection using Pearson Correlation improves classification accuracy by removing weakly correlated and irrelevant features. This reduces noise in the dataset, improves model learning, and enhances prediction reliability.

Thus, the accuracy comparison confirms that the proposed feature selection strategy significantly boosts model performance, and Random Forest emerges as the most reliable classifier for final AQI category prediction.

4.3 Precision, Recall, and F1-Score Comparison

Precision, Recall, and F1-score are important performance metrics used to evaluate the effectiveness of classification models, especially in multiclass classification problems such as AQI category prediction. While accuracy provides the overall correctness of the model, these metrics offer deeper insight into class-wise prediction quality and model reliability.

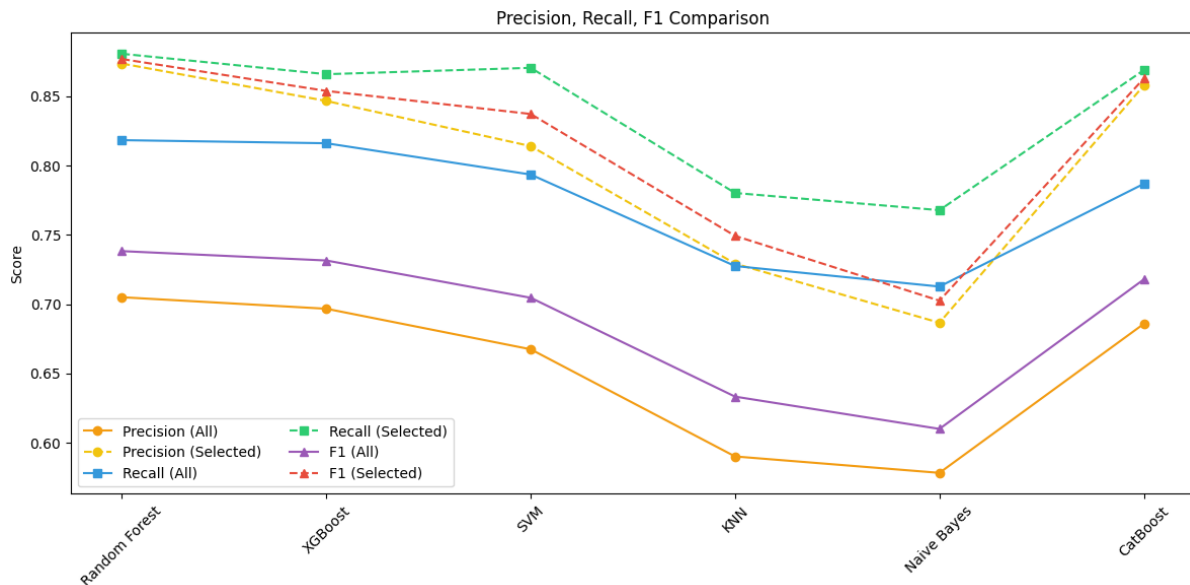


Fig. 1. System Architecture

Fig. X shows the comparative analysis of Precision, Recall, and F1-score for all six machine learning models using both all features and selected features. The solid lines represent performance using all features, while the dashed lines represent performance after feature selection using Pearson Correlation.

It is clearly observed that all models show significant improvement after selecting highly correlated features. For Random Forest, precision increases from approximately 0.70 to 0.87, recall improves from nearly 0.82 to 0.88, and the F1-score rises from about 0.74 to 0.88, indicating highly balanced and reliable classification performance.

Similarly, XGBoost demonstrates strong improvement, where precision increases from around 0.69 to 0.85, recall improves from nearly 0.82 to 0.86, and F1-score rises from approximately 0.73 to 0.85. CatBoost also performs exceptionally well, with precision improving from around 0.69 to 0.86, recall increasing from approximately 0.79 to 0.87, and F1-score reaching nearly 0.86, making it one of the top-performing models.

For SVM, precision improves from approximately 0.67 to 0.81, recall increases from around 0.79 to 0.87, and F1-score rises from nearly 0.70 to 0.84, showing that feature selection significantly enhances its predictive capability.

On the other hand, KNN and Naïve Bayes show comparatively lower performance. KNN improves from precision 0.59 to 0.73, recall 0.73 to 0.78, and F1-score 0.63 to 0.75, while Naïve Bayes improves from precision 0.58 to 0.69, recall 0.71 to 0.77, and F1-score 0.61 to 0.70. Although improved, these models still remain less effective than ensemble-based classifiers.

Among all models, Random Forest achieves the best overall balance between precision, recall, and F1-score, followed closely by CatBoost and XGBoost. This indicates that Random Forest not only predicts

correctly with high confidence but also minimizes both false positives and false negatives more effectively than other models.

The results confirm that feature selection significantly improves model robustness by removing irrelevant and weakly correlated features. This leads to better class discrimination, improved learning efficiency, and stronger predictive performance.

Therefore, based on Precision, Recall, and F1-score comparison, Random Forest is identified as the most reliable and efficient model for final AQI category prediction, providing the best balance between correctness, sensitivity, and overall classification quality.

4.4 Accuracy vs Training Time Comparison

Accuracy and training time are two important parameters used to evaluate the practical performance of machine learning models. While accuracy measures how correctly the model predicts AQI categories, training time indicates the computational efficiency required to build the model. An ideal model should provide high accuracy with reasonable training time, ensuring both prediction reliability and practical usability.

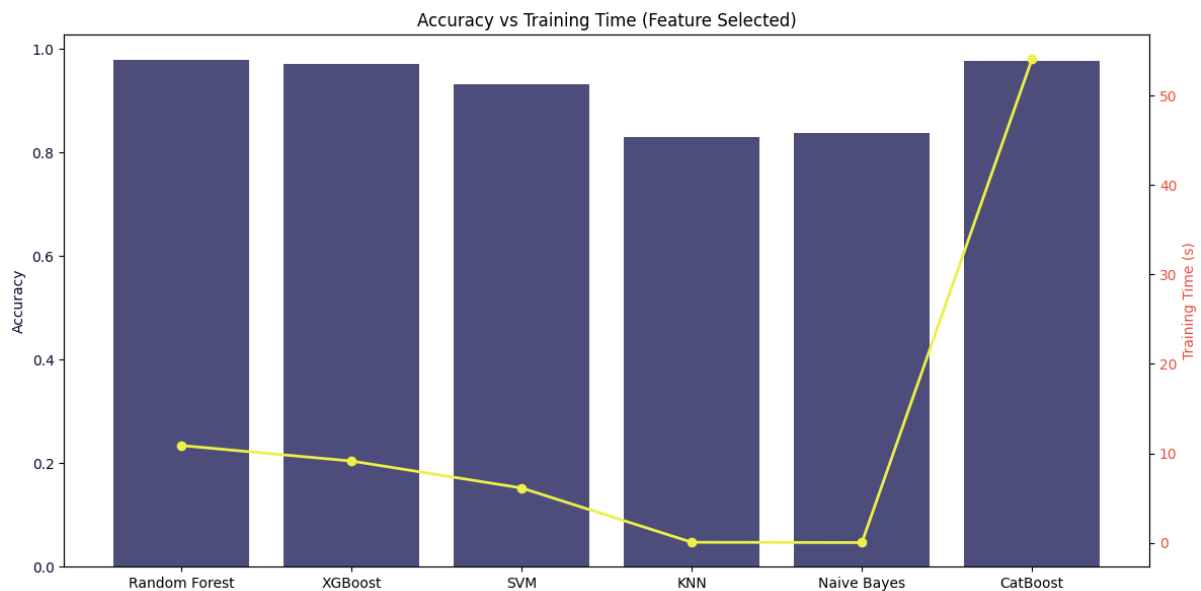


Fig. 1. System Architecture

Fig. X shows the comparison between accuracy and training time of six machine learning models using selected features. The bar graph represents the classification accuracy, while the line graph represents the corresponding training time in seconds. It can be observed that Random Forest, XGBoost, and CatBoost achieve the highest accuracy values, all close to 97–98%, whereas SVM reaches around 93%, and KNN and Naïve Bayes show comparatively lower accuracy of nearly 83–84%.

Among all models, Random Forest achieves approximately 98% accuracy with a training time of nearly 11 seconds, providing an excellent balance between performance and efficiency. XGBoost also performs strongly with about 97% accuracy and a slightly lower training time of around 9 seconds, while SVM reaches approximately 93% accuracy with nearly 6 seconds of training time. In contrast, KNN and Naïve Bayes require very little training time (close to 0–1 seconds) but their prediction accuracy remains significantly lower. Although CatBoost achieves accuracy comparable to Random Forest (around 98%), it requires the highest training time of approximately 54 seconds, making it computationally expensive.

Therefore, considering both predictive performance and computational cost, Random Forest is selected as the final model for AQI category prediction. It provides the highest accuracy with significantly lower

training time than CatBoost and stronger class-wise performance than XGBoost, making it the most efficient and reliable classifier for the proposed system.

4.5 Actual vs Predicted Output Analysis

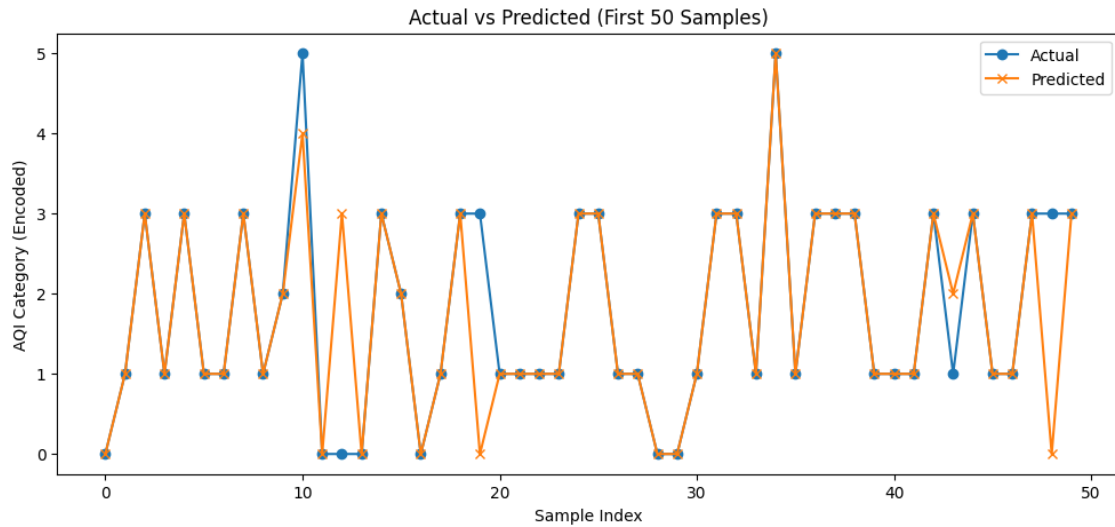


Fig. 1. System Architecture

Fig. X shows the comparison between actual AQI categories and predicted AQI categories for the first 50 samples using the best-performing model (Random Forest). The blue line represents the actual class labels, while the orange line represents the predicted values generated by the model. From the figure, it can be clearly observed that the predicted values closely follow the actual values for the majority of the samples. The two lines almost overlap throughout the plot, indicating that the model is able to accurately capture the underlying patterns in the data. Only a few minor deviations can be seen at certain sample indices (for example around indices 10–15, around index ~19, and near the end of the sequence), where the predicted class slightly differs from the actual class. However, these deviations are minimal and do not significantly impact the overall performance of the model.

Overall, the strong alignment between the actual and predicted AQI categories demonstrates that the Random Forest model has high predictive accuracy and is capable of generalizing well on unseen data. The close overlap of both lines indicates low classification error and suggests that the model effectively learns the relationships between input features and AQI categories. This visual result further supports the quantitative evaluation metrics, confirming that Random Forest is the best-performing model among all the applied algorithms.

4.5 Actual vs Predicted Output Analysis

Fig. X presents a detailed geospatial representation of Air Quality Index (AQI) severity across India, where the predicted output of the Random Forest model is mapped onto the country's state boundaries. The visualization is generated by taking multiple environmental input features—such as pollutant concentrations (PM2.5, PM10, NO₂, SO₂, CO), meteorological parameters, and temporal factors—and passing them through the trained model. The model then classifies each region into discrete AQI categories (e.g., Good, Moderate, Poor, Very Poor, Severe), which are encoded and visually represented using a color scale.

In the figure, Maharashtra is distinctly highlighted in a dark red shade, indicating that the model has classified this region under the **“Severe” AQI category**. This implies extremely high pollution levels with significant potential health risks, especially for vulnerable populations such as children, elderly individuals, and those with respiratory conditions. The remaining states are displayed in a neutral gray tone, which either represents lower AQI severity levels or regions that are not currently categorized as severe. The clear contrast between red and gray enhances visual interpretability, allowing immediate identification of critical pollution hotspots.

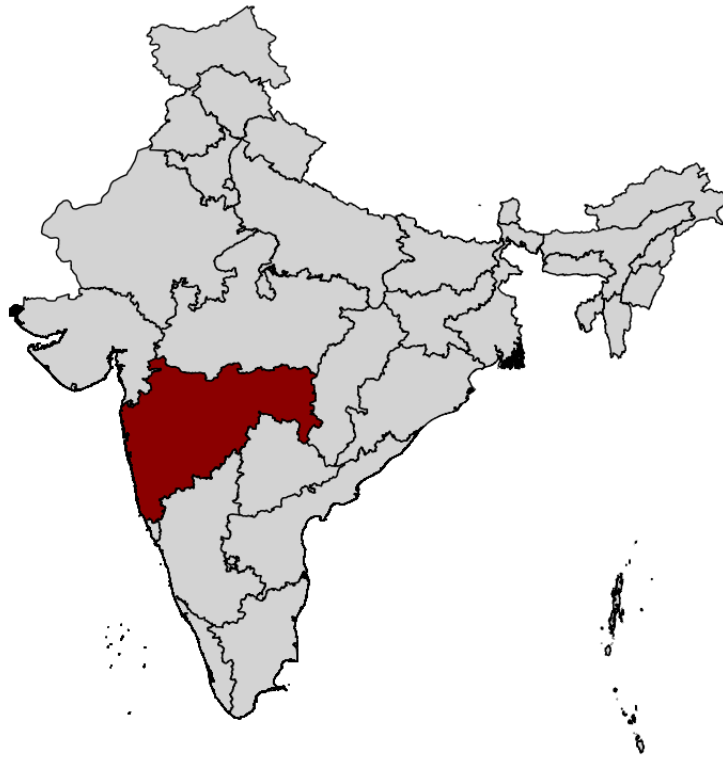


Fig. 1. System Architecture

From a technical perspective, this visualization is achieved by integrating model predictions with geospatial boundary data (such as shapefiles or GeoJSON). Each state polygon is assigned a predicted AQI label, and a corresponding color is applied based on severity. This approach transforms abstract numerical predictions into an intuitive spatial format, bridging the gap between machine learning outputs and real-world geographical insights. Additionally, such mapping enables easy comparison across regions and supports temporal extensions where AQI severity can be tracked over time.

The significance of this visualization lies in its practical applicability. It serves as a decision-support tool for government authorities, environmental monitoring agencies, and urban planners. By clearly identifying high-risk regions like Maharashtra, targeted interventions—such as emission control policies, industrial regulation, traffic management, and public health advisories—can be implemented more effectively. Furthermore, this type of visual output enhances communication with the general public, as complex model predictions are simplified into an easily understandable map format. Overall, the geospatial AQI severity map not only validates the effectiveness of the Random Forest model but also demonstrates its utility in real-world environmental monitoring and policy-making scenarios.

5. CONCLUSION AND FUTURE WORK

This study presents a comprehensive and well-structured machine learning framework for accurate prediction of Air Quality Index (AQI) categories using environmental pollutant data. A systematic pipeline is designed, beginning with data preprocessing techniques such as outlier detection using the Interquartile Range (IQR) method and feature scaling through Min-Max normalization, ensuring data consistency and stability. The issue of class imbalance is effectively handled using SMOTE, which significantly improves the representation of minority classes and enhances the model's ability to generalize across all AQI categories. Feature selection using Pearson Correlation further refines the dataset by retaining only the most relevant attributes, thereby reducing dimensionality and improving computational efficiency. A comparative evaluation of six machine learning algorithms—Random Forest, K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naïve Bayes, CatBoost, and XGBoost—demonstrates that ensemble-based methods outperform traditional models in AQI classification tasks. Among them, the Random Forest classifier achieves the best overall performance with approximately 98% accuracy, along with strong precision, recall, and F1-score values. The confusion matrix analysis highlights its minimal misclassification and strong class-wise prediction capability, confirming its robustness and reliability. Furthermore, the actual vs predicted output analysis shows a high degree of alignment, indicating that the model effectively captures underlying data patterns and maintains strong generalization on unseen data. In addition to quantitative evaluation, the integration of visualization techniques such as confusion matrices, accuracy comparison graphs, and geospatial AQI mapping significantly enhances the interpretability of the results. The geospatial representation, in particular, provides intuitive insights into regional pollution severity, making the model highly useful for environmental monitoring and policy-making. Overall, the proposed system not only achieves high predictive performance but also offers a scalable, interpretable, and practical solution for real-world air quality assessment, contributing to improved decision-making and public health management.

Future enhancements of this work can focus on developing a more advanced and real-time AQI prediction system by integrating live data streams from IoT-based air quality sensors and public APIs. This would enable continuous monitoring and dynamic prediction of air quality levels, making the system more responsive to real-world environmental changes. Incorporating time-series modeling techniques such as Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and hybrid CNN-LSTM architectures can further improve prediction accuracy by capturing temporal dependencies and seasonal variations in air pollution data. Additionally, expanding the feature set by including meteorological and environmental parameters such as temperature, humidity, wind speed, wind direction, rainfall, and solar radiation can provide deeper insights and enhance model performance. The model can also be extended to perform fine-grained predictions at the city or district level, allowing for more localized and precise air quality analysis. Integration with Geographic Information Systems (GIS) and interactive dashboards can improve visualization, enabling users to explore AQI trends and patterns more effectively. From an application perspective, deploying the model as a web or mobile-based platform can significantly increase its accessibility and usability. Features such as real-time alerts, health advisories, and pollution forecasting can be incorporated to assist both the general public and government authorities. Furthermore, optimization techniques can be applied to reduce computational cost and improve scalability for large-scale deployments. Finally, integrating automated decision-support systems that suggest pollution control measures based on predicted AQI levels can make the system more impactful in addressing environmental challenges and safeguarding public health.

REFERENCES

- [1] Pinakidou, G. Delis, D. Latinopoulos et al., "Perceptions and awareness of air quality among citizens of thessaloniki, greece," *Discover Environment*, vol. 4, p. 48, 2026.
- [2] D. Aggarwal, R. S. Shetty, S. R. Shetty et al., "Investigating the planned behavior of university students towards sustainable campus through air quality mitigation," *Discover Sustainability*, vol. 7, p. 59, 2026.
- [3] A. Taghipour, A. Saengnoree, and W. Sangmanee, "Behavioral intentions toward air purifier adoption in thai municipal offices for sustainable indoor air quality," *Discover Sustainability*, vol. 7, p. 135, 2026.
- [4] N. Kumar and Suman, "Atmospheric aerosols and urban chemistry: assessing the link between air pollution index and traffic volume in greater noida," *Spectrochimica Acta Part A*, 2026.
- [5] D. Ali, M. Singh, A. A. Khan et al., "Impact of meteorological drivers on air quality index: A case study from delhi," *Environmental Pollution*, vol. 388, p. 127324, 2026.
- [6] A. Pootheri, P. Chinnaiyan, N. Ashok, and R. Saraswathy, "Association between ambient air pollution and increased risk of respiratory diseases in vellore and ra-nipet, tamil nadu, india: A retrospective study," *Environmental Pollution*, 2025.
- [7] P. Jayaraman, G. N. Manikandan, and P. Partheeban, "Analysis of neural networks for air quality forecasting in populated areas," *Environmental Engineering Research*, 2026.
- [8] P. Fusiripong and H. Surbakti, "Aqnet adaptive mesh attention framework for spatiotemporal air quality prediction in diverse urban environments," *International Journal of Data Science and Analytics*, vol. 22, p. 34, 2026.
- [9] C. Jena, V. K. Soni, A. Kumar et al., "Evaluations of pm_{2.5} forecast during winter 2020 over india using wrf-silam model," *Atmospheric Environment*, 2026.
- [10] P. Goetzinger, S. Noy, and K. Huffstadt, "Machine learning approaches for predicting air pollution levels: a transparent, time-aware pipeline for daily aqi in indian cities," *Modeling Earth Systems and Environment*, vol. 12, p. 97, 2026.
- [11] A. K. Gond, A. Jamal, and T. Verma, "Long-term air quality forecasting in kolkata, india (2025–2047): A hybrid model using 44-year satellite data," *Journal of Atmospheric and Solar-Terrestrial Physics*, 2026.
- [12] W. Yang, W. Shang, and T. Fatima, "Hybrid tcn-bigru model for accurate air quality index prediction," *Environmental Pollution*, 2026.
- [13] W. Huang, X. Huang, Y. Zhang, and H. Zhu, "A hybrid ssa-vmd-gru model for real-time traffic-related air quality index prediction: Development and validation," *Sustainability*, vol. 17, no. 24, p. 11233, 2025.
- [14] E. Mohapatra, M. Das, and S. Rath, "Deep learning-based aqi forecasting: a cnn-lstm model with visual insights from shap-lime and pdp," *Discover Applied Sciences*, vol. 7, p. 1326, 2025.
- [15] K. M. Ordenshiya and G. K. Revathi, "Enhancing air quality index forecast with string reduction, entropy weight and similarity measure using k-means clustering for fuzzy inference system," *Engineering Applications of Computational Fluid Mechanics*, 2025.
- [16] Riechie, F. V. Ferdinand, and F. V. P. Samosir, "Comparative analysis of tree-based ensemble methods for air quality predictions in singapore using socioeconomic attributes," in *Proceedings of the 22nd International Conference on Electrical Engineering/Electronics, Computer, Telecommunications and Information Technology (ECTI-CON)*, 2025.
- [17] S. Hwang, J. Park, Y.-T. Chu, and J. Choi, "A gnn-based interpolation method for enhancing air pollution prediction based on internet of things (iot) data," *Atmospheric Environment*, 2026.
- [18] S. Dey, R. Raina, S. Mishra, A. S. Prasad, and R. Dharavath, "Greenedge ai: Sustainable federated learning for smart city air quality prediction," *Journal of Industrial Information Integration*, vol. 50, p. 101081, 2026.
- [19] V. Mathur, D. Srivastava, J. Pandey, and V. Mishra, "Soft computing techniques for atmospheric pollution and traffic emission prediction," *Environmental Modelling & Software*, vol. 197, p. 106828, 2026.
- [20] L. Arya, S. Mehrotra, C. H. Sabitha, M. Sivaparthi, and V. R. Katuri, "Real-time air quality monitoring in industrial zones using iot-powered ai systems," *International Journal of Computer Information Systems and Industrial Management Applications*, 2026.
- [21] N. D. Trong, T. D. Le, and H. Q. Nguyen, "Atmos: Iot-driven air quality monitoring and forecasting system using machine learning and fog computing," in *Communications in Computer and Information Science*, 2026.
- [22] M. U. S. Imsek, K. Kök, and S. Özdemir, "Deepfogaq: A fog-assisted decentralized air quality prediction and event detection system," *Expert Systems with Applications*, vol. 251, p. 123920, 2024.

- [23] A. Y. P. Asih, F. Yudianto, P. Triwinanto, R. S. Marjianto, T. Herlambang, and H. Arof, "Comparison of xgboost and random forest methods in predicting air pollution levels," *BAREKENG: Journal of Mathematics & Applications*, vol. 20, no. 1, pp. 0785–0796, 2026.
- [24] G. Ravindiran, K. Karthick, S. Sivarethnamohan et al., "A cluster-aware synthetic resampling and machine learning framework for multi-class air quality index classification," *Environmental Monitoring and Assessment*, vol. 198, p. 270, 2026.
- [25] M. Kallista, I. P. D. Wibawa, and S. C. Obie, "Evaluating the effectiveness of kernel extreme learning machines over conventional elm for air quality index prediction," *Barekeng*, vol. 20, no. 2, pp. 1373–1388, 2026.
- [26] S. Kasim, S. Malek, S. Cheen et al., "Machine learning-based prediction of mortality risk from air pollution-induced acute coronary syndrome in the western pacific region," *Scientific Reports*, vol. 16, p. 3486, 2026.
- [27] V. Ivanovic, M. A. J. Sujan, O. J. Mengshoel, and T. Moholdt, "Artificial intelligence methods in gestational diabetes mellitus prediction: A systematic literature review," *International Journal of Medical Informatics*, vol. 206, p. 106158, 2026.
- [28] O. Charjan, K. Gajbhiye, J. Warhade et al., "A feature engineering-driven ensemble approach for accurate aqi forecasting," *Discover Applied Sciences*, vol. 8, p. 216, 2026.
- [29] J. Kuruvilla and S. Vasudevan, "Residual stacking with temporal convolutional network and light gradient boosting for air quality index prediction," *International Journal of Intelligent Engineering and Systems*, 2026.
- [30] S. T. Mim, F. Alam, and N. Sharmin, "Air quality index prediction: Comparative study based on various hyper parameter tuning," *Iran Journal of Science*, vol. 49, pp. 1563–1586, 2025.
- [31] Y. El Mghouchi and M. T. Udristoiu, "Integrated ai framework for sustainable environmental management: Multivariate air pollution interpretation and prediction using ensemble and deep learning models," *Sustainability*, vol. 18, no. 3, p. 1457, 2026.
- [32] U. Ullah, A. Idris, and R. A. Wagan, "Aqi prediction for dry climate: A case study of quetta city using ensemble learning," *Theoretical and Applied Climatology*, vol. 156, p. 571, 2025.
- [33] T. Madan, S. Sagar, Y. Singh et al., "Hybrid deep learning model for air quality prediction and its impact on healthcare," *Scientific Reports*, vol. 16, p. 6036, 2026.
- [34] S. Kalaiselvi, V. Anitha, V. Manimaran et al., "Air quality prediction using multi-source remote sensing data integration with hybrid deep learning framework," *Scientific Reports*, vol. 16, p. 2688, 2026.
- [35] B. Li, Y. Cui, L. Li, and Y. Li, "Advanced prediction of mine airflow based on woa-optimized vmd and a deep learning hybrid model," *Tunnelling and Underground Space Technology*, 2026.
- [36] H. S. Iyer, M. Perlstein, S. A. Joseph, C. Roscoe, N. Zeinomar, A. P. Cole, F. R. Schumacher, J. Beebe-Dimmer, J. J. Ruterbusch, C. L. Wiggins, A. M. Stroup, J. Schwartz, J. E. Hart, T. R. Rebbeck, and F. Laden, "Associations of fine particulate matter pollution components with overall, prostate cancer, and cardiovascular disease mortality in men with prostate cancer: A cohort study," *Environmental Research*, vol. 294, p. 123909, 2026.
- [37] S. S. Subbiah, S. K. Paramasivan, and M. Thangavel, "Prediction of particulate matter pm2.5 using bidirectional gated recurrent unit with feature selection," *Global Nest Journal*, 2024.